

Team Name: FindMyMeal

Group members: Benaya Berhane, Bazen Mekonen,
Muhammad Wajahat, Hunnain Arsalan, Ishan Giri,
Michael Glass

Class Name: Structured Systems Analysis and Design

Class Code: IS436

Deliverable Name: Deliverable 1 - "System Request"

Date: 9/29/2024

Systems Request

Project Sponsor: Professor Shiva Sharma and the class of IS436 Fall 2024

Business Need: To help grow our business into new avenues with revenue. This will also help our new teams grow their team working skills by going through the SDLC, implementing a new system, and seeing where within this they struggle.

Business Requirements: This system will provide users ways to find meals by taking in the ingredients that they have and give them recipes. This will include prices of ingredients if you want to make a new recipe, what you can make from ingredients you have, and even different cuisines that may be favored.

Business Value: We will see a huge improvement within our team, especially with their problem solving skills. We will also see more people going towards our site and we may even see others using our website showing how important this class is and how intelligent our instructor is.

Special Issues: This must be done before the end of UMBC's Fall 2024 semester. We will also not be reimbursed for any money spent and can't have our instructor or others students work on this as well.

Feasibility Analysis

For our feasibility analysis we are looking at three things: Can we build it(Technical), Is this really worth doing(Economic), and Will the organization accept this if we go ahead(Organizational). We will be starting with the Technical analysis, going to the Economic analysis, and finally finishing with the Organizational analysis.

For the technical analysis, we are asking if we can do this. I am saying that we can. First, we have the spoonacular api. This api allows us to classify recipes based on their diets, allergies, cuisines, and the ingredients they have. It gives the total cost and the cost per serving, with additional nutritional information. Along with that we will create a database that users can be saved into, having all their preferences ready for the next time they log in. We have familiarity with creating a website, through the use of HTML/CSS/JavaScript and have database experience using oracle. We also have some backend experience, but this is more self taught than class/formally taught. So for some this may be a bit newer. The team is made up of 6 members. This is a standalone system so we are not worrying about compatibility.

For the economic analysis, we are asking if we should even build this. I am saying yes to this question. First, the cost is negligible as the API spoonacular allows for us to use it and as long as there are no real formal calls everything should be fine. If there are calls then they are extremely cheap. The return on investment would just be a working system that we can put into production. If we employ a web interface that is clean both on a phone, laptop, and computer we will have many customers coming back. We can even begin to use the data we gather for other purposes. Some of the costs will be the team members salaries(grades), development testing, api call costs, and storage fees. The benefits will be an increase in salary(getting an A), increased users, and increased data.

For the organizational feasibility, we are asking if the organization will be ok with this project. I fully believe they will, and here is why. First we are proving to the backers that we are a competent team that can create useful applications. Next we will be allowing the backers a system that goes into a market no other team is heading towards. Finally, we are allowing for the class to be involved in a project that can be used in their day to day lives. This perfectly aligns with the backers as we are showcasing what we can do and implementing software that can be used by others.

Cost Benefit Analysis

2024: Basic testing tools, development time, API subscription and hosting setup (50\$ to 150\$)

Benefits:

Beta version of the prototype to evaluate and early user feedback.

Benefit of improved team experience.

Early exposure to real-world development.

2025: Further use of API usage (100\$).

Continued ongoing server hosting and database storage (150\$).

Commitment from the team for maintenance and bug fixes.

Benefits:

We can grow user base with very valuable information and data on user preferences and behaviors

High chance of potential partnership with food delivery services, grocery stores etc.

Improved system reliability and performance through ongoing updates

2026: Scaling server and API cost (200\$)

Updated features and expansion of system (200\$)

Benefits:

Continued user growth

Increased platform engagement

Potential expansion to new market

Monetization potential (Ads)

Potential Total costs in 3 years - **650\$**

Project Methodology:

The **V-Model methodology** has been selected for this project as it offers a clear, structured development approach, emphasizing the relationship between testing and development stages. This model ensures issues are identified early in the process, reducing the risk of discovering defects later.

Phases of V-Model:

- **Requirement Phase:** Identify user needs and system requirements.
- **Design Phase:** Develop high-level and detailed designs for the system.
- **Implementation Phase:** Code and develop the system based on designs and requirements.

- **Testing Phase:** Conduct thorough testing (unit, integration, system testing) as each module is completed.
- **Deployment Phase:** Deploy the system for beta testing and gather user feedback for further improvements.

Justification: The V-Model ensures each component is inspected before progressing to the next phase, reducing the risk of major issues in later stages. Unlike Agile, the V-Model's linear, well-defined progression fits the academic structure of our project, where documentation and stepwise verification are critical.

Project work plan

Milestone 1: Feasibility Analysis Completion

- **Description:** Complete the feasibility analysis including Technical, Economic, and Organizational aspects.
- **Start Date:** 9/30/2024
- **End Date:** 10/7/2024
- **Deliverables:** Documented feasibility analysis report.

Milestone 2: Requirements Gathering

- **Description:** Identify user needs and document system requirements.
- **Start Date:** 10/8/2024
- **End Date:** 10/14/2024
- **Deliverables:** Requirements specification document.

Milestone 3: System Design

- **Description:** Develop high-level and detailed design of the system.
- **Start Date:** 10/15/2024
- **End Date:** 10/28/2024
- **Deliverables:** System design document including architecture, database schema, and user interface design.

Milestone 4: Development Phase

- **Description:** Implement the system based on the design and requirements.
- **Start Date:** 10/29/2024
- **End Date:** 11/25/2024

- **Deliverables:** Fully functional codebase.

Milestone 5: Testing Phase

- **Description:** Conduct thorough testing (unit, integration, system testing).
- **Start Date:** 11/26/2024
- **End Date:** 12/10/2024
- **Deliverables:** Testing report including identified issues and fixes.

Milestone 6: Deployment

- **Description:** Deploy the system for beta testing and gather user feedback.
- **Start Date:** 12/11/2024
- **End Date:** 12/15/2024
- **Deliverables:** Deployed application and user feedback report.

Milestone 7: Final Presentation and Report

- **Description:** Prepare final project report and presentation for stakeholders.
- **Start Date:** 12/16/2024
- **End Date:** 12/20/2024
- **Deliverables:** Final report and presentation materials.

Resource Allocation

- **Technical Resources:** Spoonacular API, Oracle Database, HTML/CSS/JavaScript for front-end development.
- **Human Resources:** Six team members, rotating roles to ensure project management responsibilities are shared.

Risk Management

- **API Limitations:** We will closely monitor API usage and conduct stress testing to ensure scalability.
- **Team Availability:** Weekly meetings and a shared project management board (GitHub) will ensure everyone is aligned. Each team member has a backup in case of unexpected absences.
- **Technical Challenges:** Code reviews will help identify and solve issues early, while continuous integration will ensure code is always deployable.
- **Feedback Delays:** We will engage with beta users early to gather feedback and iterate on the application before the final release.

Communication Plan

- **Weekly Meetings:** Mondays at 3 PM.
- **Updates:** Weekly on GitHub.
- **Tools:** Slack for communication, GitHub for project management.

Team Information

Team Name: FindMyMeal

E-mail: Hw71819@umbc.edu

Project Contact Person: Muhammad Wajahat (Project Manager)/ Bazen Mekonen (Backend Developer)

Time Set Aside to Meet Outside of Class: Mondays and Wednesdays at 4 pm

GitHub Project Link: <https://github.com/users/bazenmekonen/projects/4>

Bios of Team Members:

- **Muhammad Wajahat (Project Manager):** Provides leadership for the team by coordinating tasks, ensuring that milestones are met on time, and maintaining a clear communication flow among all members.
- **Benaya Berhane (Database Developer):** Specializes in database design and management, focusing on creating efficient and scalable databases for storing user data and preferences.
- **Bazen Mekonen (Backend Developer):** Responsible for building and maintaining the server-side logic, integrating the database with the front-end, and ensuring seamless data processing.
- **Hunnain Arsalan (Frontend Developer):** Designs and develops the user interface, ensuring a user-friendly experience with responsive and intuitive designs.
- **Ishan Giri (Networking Specialist):** Focuses on the networking aspects of the project, ensuring that the system remains secure and the API interactions are optimized.
- **Michael Glass (Database Developer, Testing):** Assists with database development while also conducting rigorous testing to ensure the system's functionality and performance meet the required standards.